David Camilo Navarro Saiz

Senior AI Engineer | Production Agentic Systems, LLM Evaluation & AI Platforms

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Summary

Senior AI Engineer with 7+ years architecting and delivering AI systems, from industrial computer vision and applied machine learning to production LLM agents and enterprise AI platforms. Architected multilingual, policy-governed LangGraph workflows with layered safeguards and verified business actions, while building end-to-end agent evaluation infrastructure spanning synthetic datasets, RAG metrics, and experiment observability. Combines hands-on Python/TypeScript engineering, MLOps, and AWS/Azure cloud expertise with proven technical leadership, including leading a 10-engineer team and delivering 30+ industrial machine-vision projects.

Skills

Programming & Backend	Python, TypeScript, JavaScript, SQL, C++, FastAPI, Node.js, React, Next.js, PostgreSQL
Agentic AI & LLMs	LangGraph, LangChain, PydanticAI, LlamaIndex, Retrieval-Augmented Generation (RAG), MCP, AI Agent orchestration
Machine Learning & Evaluation	PyTorch, TensorFlow, Computer Vision, RAGAS, DeepEval, MLflow
AWS & Azure	Amazon Bedrock, AgentCore, ECS/Fargate, RDS, AKS, Azure Container Apps, App Service, Azure AI Search, Azure AI Speech
DevOps & MLOps	Docker, Kubernetes, Terraform, GitHub Actions, CI/CD, DVC, CVAT, Git
Languages	English (Fluent), Spanish (Native)

Technical Experience

Svitla Systems

Aug 2025 — Present

Sr. ML/AI Engineer

Roadie — Agentic AI Platform

Nov 2025 — Present

- Architected and delivered a multilingual, policy-governed LangGraph pickup-support agent to production, combining natural-language interaction with deterministic business rules, controlled state transitions, and authorized tool execution.
- Designed layered safeguards for sensitive agent operations, integrating eligibility checks, confirmation gates, tool-result verification, and LLM response adjudication to reduce the risk of unsupported action-completion claims.
- Engineered a shared agent-runtime reliability layer with structured execution outcomes, operation-level timeouts, retry classification, and predictable fallback behavior during tool and provider failures.
- Built an end-to-end agent and RAG evaluation framework combining synthetic QA generation, configurable experiments, RAGAS, DeepEval, retrieval ranking metrics, and experiment tracking to assess agent quality and diagnose retrieval and response failures.
- Developed observability and human-review workflows using execution traces, retrieved context, Opik experiments, and scenario-testing interfaces.

360Training — Computer Vision & ML

Aug 2025 — Nov 2025

- Engineered a multi-signal proctoring pipeline combining face detection, gaze estimation, head-pose analysis, and temporal behavioral features, integrating ML inference and heuristic services into application workflows.
- Built reproducible ML experimentation pipelines, benchmarking XGBoost, LightGBM, Random Forest, logistic regression, and neural models using CVAT annotation, DVC dataset versioning, and MLflow tracking.

Kmeleon

Jan 2025 — Aug 2025

Generative AI Engineer

- Architected a configurable multi-agent platform using LangChain and PydanticAI, enabling enterprise users to create AI assistants with conversational memory, tool execution, and collaborative workflows.
- Designed Model Context Protocol (MCP) integrations connecting AI agents with external tools and enterprise services, including Microsoft Teams, through reusable tool interfaces.
- Integrated retrieval-augmented generation, agent configuration, and Azure-oriented infrastructure into modular AI applications, supporting extensible workflows and maintainable LLMOps practices.

Southdesk

Apr 2024 — Jan 2025

Machine Learning Engineer

- Developed RAG-powered applications combining LLMs, vector retrieval, and GoHighLevel CRM integrations to automate business knowledge access, customer onboarding, and personalized communication workflows.
- Engineered web-scraping and data-extraction pipelines integrated with CRM automation, transforming prospect information into structured data for lead generation and outreach.

El Sol — Neighborhood Educational Center

Jun 2023 — Apr 2024

Machine Learning Team Leader

- Led end-to-end engineering of conversational AI and document-intelligence applications using FastAPI, React/TypeScript, Node.js, and LangChain, integrating multi-format document ingestion, RAG, retrieval filtering, and reranking.
- Developed AI-powered automation workflows leveraging Azure AI Speech for voice-assisted data collection, intelligent form completion, personalized reporting, and document-based recommendations.
- Architected and deployed containerized AI services using Azure Kubernetes Service (AKS), Azure App Service, and Azure Container Apps, integrating CI/CD pipelines for repeatable application delivery and updates.

INTECOL SAS

May 2019 — Apr 2023

Computer Vision Engineer & Team Leader

- Led a team of 10 engineers delivering 30+ industrial computer-vision projects, overseeing technical architecture, project planning, client requirements, budgets, and cross-functional delivery.
- Designed and integrated machine-vision systems using CNNs, YOLO, PyTorch, TensorFlow, Python, and C++, translating deep-learning models into operational industrial automation applications.
- Developed computer-vision solutions for object detection and human-behavior analysis, integrating model inference into desktop applications and real-world monitoring systems.

Education

B.Sc. in Physical Engineering
Universidad Nacional de Colombia, Medellin

MLOps & Data Science: MLOps Specialization (Duke University / Coursera); DS4A – Data Science for All (Correlation One / MinTIC).

Deep Learning & NLP: DeepLearning.AI coursework in Deep Learning, TensorFlow and GANs; Natural Language Processing Specialization (Coursera).

Generative AI & Agents: Specialized coursework in LangGraph, LLM Engineering, LangChain, and AI agent development.

Current Learning Focus: AWS Solutions Architect – Associate certification preparation and advanced cloud-based AI architecture.

Selected Projects & Technical Initiatives

Parser AI (ParserDigital) – Enterprise AI Platform & Coding-Agent Evaluation

AI Coding-Tool Evaluation: Designed and implemented a reproducible benchmarking workbench for Cursor, Claude Code, Codex, GitHub Copilot, and Gravity. Standardized full-stack development tasks, versioned prompts, and 37 validation checks, capturing code diffs, execution transcripts, CI results, security checks, timing, token usage, and human interventions. Combined deterministic validation with LLM-as-a-Judge and human scoring rubrics to assess functional correctness, code quality, and agent autonomy, supporting comparative analysis through a leaderboard.

Enterprise AI Platform & AWS: Engineered infrastructure for a LibreChat-based enterprise AI platform, validating RAG persistence migration to Amazon RDS PostgreSQL/pgvector with database access controls and S3 integration. Built authenticated AI-generated application previews using Sandpack, ECS/Fargate, HMAC-based authorization, and ALB routing.

Professional Cases – Multimodal AI Case Simulation & Assessment

Agentic Case Generation: Architecting a scalable, domain-agnostic platform that transforms professional documentation, images, and specialized knowledge into realistic, interactive case studies. Designing agent-driven research and retrieval workflows to generate context-rich scenarios that challenge users to apply domain expertise, critical thinking, and real-world decision-making.

Knowledge Diversity & AI Assessment: Developing semantic clustering and knowledge-coverage strategies to diversify case topics, complexity, and reasoning challenges while reducing repetitive scenarios. Building rubric-based LLM assessment to evaluate user reasoning, identify knowledge gaps, and deliver personalized, actionable feedback. The platform is designed around Next.js, FastAPI, PostgreSQL/pgvector, and Redis.

Applied AI Research – Computational Imaging & Deep Learning

Contributed to research at Universidad Nacional de Colombia exploring CNN-based holographic image reconstruction and computational imaging. Led the development of an experimental thermography-based injury detection system. For COVID-RX, developed and compared PyTorch CNN architectures, including ResNet, DenseNet, and EfficientNet, using 36,000+ publicly sourced chest X-ray images for COVID-19 and pneumonia classification experiments.

Pupil Tracking Research – SICC 2019

Presented applied computer-vision research on automated pupil localization and tracking using CNN-based image analysis and machine vision techniques. Explored extracting eye-position and movement information from visual data, with potential applications in gaze-based human-computer interaction and educational technologies.

Computer Vision Co-Instructor – Universidad Nacional de Colombia

Collaborated with INTECOL SAS to design and deliver technical training in image processing, deep learning, and practical computer-vision applications, connecting academic concepts with real-world engineering challenges.